Vaishnav Sreekumar

+91 9037350803 | vaishnavsreekumar301@gmail.com | GitHub | LinkedIn

SUMMARY

Computer Science undergraduate passionate about AI-assisted backend systems, cloud infrastructure, distributed systems, and scalable software engineering using Python, AWS, REST APIs, and LLMs.

SKILLS

Languages: Python, JavaScript, Go, Java, SQL
AI & Machine Learning: LLMs, OpenAI API, Prompt Engineering, Pydantic, Structured AI Outputs
Backend Engineering: REST APIs, Distributed Systems, Microservices, Authentication, RBAC
Cloud & DevOps: AWS (Lambda, EventBridge, DynamoDB, IAM, CloudWatch, Organizations), Terraform, Docker, Kubernetes
Data & Observability: PostgreSQL, Redis, Apache Kafka, Prometheus, Grafana, ELK Stack
Software Engineering: Git, GitHub, Linux, Unit Testing, CI/CD, Dependency Injection

OPEN SOURCE CONTRIBUTIONS

- **Kerno (GSSoC 2026)** *Go Cloud Native Linux*
  - Contributed production-quality features to an open-source cloud observability platform by implementing the `kerno doctor -list-rules` CLI capability and improving rule discovery workflows.
  - Designed and implemented automated unit tests, collaborated with maintainers through iterative code reviews, and validated changes using CI pipelines and pull requests.
- **Kubernetes Ecosystem – kubernetes-network-drivers (#24)** *Go Kubernetes Cloud Native*
  - Contributed improvements to Kubernetes Dynamic Resource Allocation (DRA) authorization workflows, strengthening RBAC-based access control for cloud-native infrastructure.
  - Gained experience contributing to large-scale upstream open-source projects by following collaborative development, review, and contribution workflows.

SELECTED PROJECTS

SentinelFinOps: AI-Assisted Cloud Governance & FinOps Platform **GitHub**
Python AWS OpenAI API Terraform Lambda DynamoDB EventBridge Pydantic

- Architected an enterprise-grade AI-assisted cloud governance platform that discovers idle AWS resources, generates explainable infrastructure recommendations using LLMs, and enforces deterministic policy validation before remediation.
- Designed a modular AI runtime with provider abstraction, context engineering, prompt management, schema validation, telemetry, and provider-agnostic execution supporting both production and mock AI providers.
- Built an event-driven AWS architecture using Lambda, EventBridge, CloudWatch, IAM, DynamoDB, and Terraform with distributed remediation locks using DynamoDB Conditional Writes to eliminate concurrent execution race conditions.
- Developed production-grade software following clean architecture principles with dependency injection, immutable Pydantic contracts, automated testing (116+ unit/integration tests), and extensible cloud-native components.

SentinelX: Multi-Tenant SIEM & Observability Platform **GitHub**
Python Apache Kafka Prometheus Grafana ELK Stack Linux

- Architected a multi-tenant security observability platform for real-time log ingestion, monitoring, and visualization using Apache Kafka, Prometheus, Grafana, and the ELK Stack.
- Designed scalable event-driven detection pipelines for SSH brute-force attacks, reconnaissance, suspicious authentication activity, and malicious web requests with automated alert generation.

Zero Trust Access Platform **GitHub**
Go PostgreSQL Docker JWT REST APIs

- Designed and developed a secure backend access platform in Go implementing RESTful APIs, JWT authentication, Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA), and identity-aware authorization following Zero Trust principles.
- Built modular backend services with PostgreSQL and Docker, emphasizing clean architecture, secure coding practices, and scalable authentication workflows.

TECHNICAL WRITING

How I Prevented Race Conditions in AWS Lambda Using DynamoDB Conditional Writes **Read Article**
Published a deep-dive technical article explaining the design and implementation of distributed locking using DynamoDB Conditional Writes, optimistic concurrency control, and fault-tolerant AWS Lambda execution for production-grade cloud systems.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner (CLF-C02)** Amazon Web Services
- **IBM Machine Learning with Python** IBM | Coursera

EDUCATION

Vellore Institute of Technology, Bhopal Sep 2023 – July 2027
Bachelor of Technology (B.Tech), Computer Science and Engineering

- CGPA: **8.15 / 10**